

Roton-like Reconstruction of the Emergent Photon in Quantum Spin Ice

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The emergent photon of quantum spin ice is well understood at long wavelength, but its fate at lattice momenta—where the compact spin- $\frac{1}{2}$ gauge theory has no small parameter—has never been resolved eigenstate by eigenstate. We compute the operator-resolved $S^{zz}(\mathbf{q}, \omega)$ of the pure pyrochlore ring-exchange model on 48-, 64-, 72-, and 96-site clusters, the largest a constrained sector of dimension 1.47×10^8 , and find five things. First, near the zone-boundary L point the spectrum reconstructs at order one: a dominant level at $0.90\text{--}1.07K$ takes 50–54% of the longitudinal weight at roughly half the Gaussian-photon energy, while a second level remains photon-like at $1.65\text{--}1.81K$. Second, the low mode is microscopically distinct, gaining ring flippability where nonreconstructing momentum classes at identical $|\mathbf{q}|$ lose it. Third, the effect is stable across four clusters, three cluster shapes, and a factor 1.6×10^4 in Hilbert-space dimension, with a scale-free branch ratio $\omega_{\text{low}}/\omega_{\text{high}} = 0.55\text{--}0.59$. Fourth, where it occurs is predicted before any excited state is computed by the Feynman ratio $f_1(\mathbf{q})/S(\mathbf{q})$ of oscillator strength to static structure factor, which discriminates correctly between inequivalent L classes on two clusters and locates a second soft momentum away from L ; sign-free Green’s-function Monte Carlo carries this ground-state diagnostic to complete momentum grids of 2048 spins, where the L -centered minimum persists with under two percent drift. Fifth, the phenomenology is that of a lattice roton and predicts a two-feature zone-boundary response—split by nearly a factor of two, weighted two-to-one, both ratios independent of the overall energy scale—that no single renormalized photon pole can produce.

Introduction.—Quantum spin ice realizes a compact $U(1)$ lattice gauge theory in a magnetic insulator. In the two-in–two-out manifold of the pyrochlore lattice, virtual exchange resonates the six spins around an elementary hexagon, and the resulting deconfined phase supports a gapless emergent photon together with gapped electric and magnetic charges [1–6]. That long-wavelength description is quantitatively established and motivates the search for candidate materials [7–16].

At lattice momenta the same theory is a different problem. The electric field is microscopically saturated, $E_\ell = S_\ell^z = \pm \frac{1}{2}$, so the usual electric term $\sum_\ell E_\ell^2$ is a constant inside the constrained manifold and the ring dynamics constitute the entire nonconstant Hamiltonian. The photon stiffness is generated, not bare, and the emergent fine-structure constant is large [17]. Nothing in this construction restricts the theory to small $\mathbf{q}$: the ring model is a lattice Hamiltonian with excitations throughout its Brillouin zone, and the short-wavelength regime is precisely where the interactions are strongest and least explored.

Existing numerics stop short of that regime in a specific sense. Gaussian and semiclassical treatments expand about small flux and are controlled in the infrared, though they already predict substantial anharmonic renormalization and an interaction-induced splitting of the transverse polarizations [7, 18–20]. Quantum Monte Carlo with analytic continuation resolves broadened lineshapes and finds zone-boundary weight beyond Gaussian electrodynamics, but not individual eigenstates or their matrix elements [21]. Constrained-space exact diagonalization has reached 96 spins, organized around ground-state and low-energy electrodynamic parameters

rather than the full response [17]. What has not been done is the direct question: at large $|\mathbf{q}|$, *which* eigenstates of the compact theory carry the spin response, and how much.

We answer it, and the answer is not a renormalized photon. Our punchlines are: (i) near the zone-boundary L point one collective level takes over half of the longitudinal spectral weight at roughly half the Gaussian-photon energy, while a second level remains photon-like nearly a factor of two above it; (ii) the low mode is microscopically distinct, gaining ring flippability where nonreconstructing momentum classes at identical $|\mathbf{q}|$ lose it; (iii) the effect is stable across four clusters, three shapes, and four orders of magnitude in Hilbert-space dimension, with a scale-free branch ratio $0.55\text{--}0.59$; (iv) its location is predicted from ground-state data alone through the Feynman ratio f_1/S , which we verify as a forward diagnostic three times and carry to 2048 spins by sign-free Monte Carlo; and (v) the resulting two-feature zone-boundary response is an experimentally distinguishable alternative to a single renormalized photon pole.

Model, observable, and method.—We study the standard ring Hamiltonian in the spin-ice manifold [1, 8], $H = -K \sum_p (W_p + W_p^\dagger)$ with $W_p = S_1^+ S_2^- S_3^+ S_4^- S_5^+ S_6^-$ summed over contractible elementary hexagons; in gauge language $W_p = e^{iB_p}$ and H is the compact magnetic energy $-2K \sum_p \cos B_p$. Because S^z is the emergent electric field, the natural probe is the sublattice-summed longitudinal response

$$S^{zz}(\mathbf{q}, \omega) = \frac{1}{N} \sum_{\mu, n} |\langle n | S_\mu^z(\mathbf{q}) | 0 \rangle|^2 \delta(\omega - \omega_n), \quad (1)$$

with $S_\mu^z(\mathbf{q}) = \sum_{i \in \mu} e^{-i\mathbf{q} \cdot \mathbf{r}_i} S_i^z$. All spectral fractions be-

low are normalized within Eq. (1) at fixed $\mathbf{q}$. This is the theoretical pseudospin response, not a material-specific neutron cross section, which additionally involves coherent sublattice interference, the local easy axes, the g tensor, form factors, and the polarization projector; we return to this point and treat it fully in the Supplemental Material.

We evaluate Eq. (1) by complete diagonalization at 48 sites and by Lanczos continued fractions at 64, 72, and 96, the last in a winding sector of dimension 146 758 352. Winding sectors are certified exhaustively at 48 and 72 sites—every one of the 155 and 310 sectors diagonalized, with margins of $0.99K$ —and by the twelve largest of 519 candidates at 96. Recomputing the 96-site spectra at Krylov dimension 200 rather than 120 leaves the dominant pole energies and residues unchanged to machine precision. Cluster construction, momentum certification, gates, and convergence are detailed in the Supplemental Material.

An unexpected low-energy mode.—Figure 1 shows the operator-resolved spectrum along $\Gamma \rightarrow L \rightarrow (0, \frac{1}{2}, \frac{1}{2}) \rightarrow X \rightarrow \Gamma$. Near Γ the dominant response tracks the expected long-wavelength photon. Near L it does not. The spectrum reorganizes into two dominant levels, and the lower one takes most of the weight (Table I). Fixing the single Gaussian energy scale from the long-wavelength spectrum places the Gaussian photon at L near $2.03K$; the low mode sits at roughly half of that, while its partner sits at 81–89%, the magnitude of renormalization an ordinarily interacting photon exhibits.

Throughout we call these two levels simply the *lower* and *upper branch*. The terminology is operational: at each momentum the S^z probe lives in a two-dimensional transverse subspace of the gauge field, and the two dominant levels couple through nearly orthogonal directions of that subspace, so they are the two independent collective features the probe can create there. No adiabatic connection to a free photon is implied by either name; with E^2 a c -number there is no coupling constant to turn off, and hence no such connection to construct.

What the mode does that a photon does not.—Two

of its properties are already in Table I: it is weight-dominant, carrying two to three times the response of its partner, and it is anomalously soft while that partner is not. A third is microscopic and uses no fitted energy scale. Adding the standard Rokhsar–Kivelson term, $H(V) = H + V \sum_p n_p^{\text{flip}}$ with n_p^{flip} the projector onto a flippable alternating hexagon, and applying Hellmann–Feynman at the physical point $V = 0$ gives

$$\left. \frac{d\omega_n}{dV} \right|_{V=0} = \left\langle n \left| \sum_p n_p^{\text{flip}} \right| n \right\rangle - \left\langle 0 \left| \sum_p n_p^{\text{flip}} \right| 0 \right\rangle, \quad (2)$$

an exact state-resolved census of the flippable hexagons the excitation adds or removes. The ground state maximizes ring resonance, so a typical excitation reduces flippability; averaged over a sector the census is exactly $\text{nnz}(H)/D - \langle n^{\text{flip}} \rangle_0 < 0$, and at 48 sites only 58 of 9104 eigenstates are positive.

The lower branch is positive at every size (Table I): it rearranges spins so that *more* hexagons could flip while the phases of those ring processes are less coherent, paying its energy in resonance rather than in the ice rule. The sharpest form of the statement is a comparison made within a single cluster. The anisotropic 64- and 96-site boxes split the L star into two inequivalent classes at identical $|\mathbf{q}|$ and identical Gaussian energy; the class that reconstructs has census $+0.338$ and $+0.259$, while the class that does not has -0.166 and -0.015 , -0.088 (Fig. 2). We claim no general sign rule: the upper branch’s census is negative at 48 and 64 but positive at 72 and 96, and the census magnitudes are the least size-stable of our diagnostics. It is the same-momentum class contrast, which requires neither a fitted scale nor a comparison across sizes, that identifies the reconstruction microscopically.

Robustness.—The three cluster shapes $2 \times 2 \times n$, $2 \times 3 \times 3$, and $2 \times 3 \times 4$ sample different boundary identifications, and the reconstruction survives all of them. The most robust measure is the scale-free branch ratio, 0.55–0.59 across a factor 1.6×10^4 in sector dimension, followed by the weight fraction, 50–54%; neither requires the Gaussian fit (Fig. 3). Absolute energies drift by about ten percent, as expected at these sizes, and the census magnitude drifts more. Not everything is stable, and we say which: the polarization splitting at X falls from 42% at 48 sites to 6% at 64 and we draw no conclusion from it.

Why there: the Feynman ratio.—The location of the reconstruction is not accidental, and it can be anticipated without computing a single excited state. Two moments of Eq. (1) are ground-state quantities. The zeroth is the equal-time structure factor $S(\mathbf{q})$, which measures how much static correlation at wavevector $\mathbf{q}$ the ground state already contains. The first, obtained from the double

TABLE I. The reconstruction across all four clusters. Weights are fractions of S^{zz} at the same momentum; $d\omega/dV$ is the flippability census of Eq. (2).

sites	48	64	72	96
sector dimension	9104	2.5×10^5	9.7×10^5	1.5×10^8
ω_{low}/K	1.024	0.954	1.068	0.904
weight	50.8%	50.0%	54.0%	54.4%
ω_{high}/K	1.730	1.652	1.814	1.647
weight	32.3%	31.3%	17.8%	22.5%
$\omega_{\text{low}}/\omega_{\text{high}}$	0.592	0.578	0.589	0.549
$d\omega_{\text{low}}/dV$	+0.339	+0.338	+0.134	+0.259

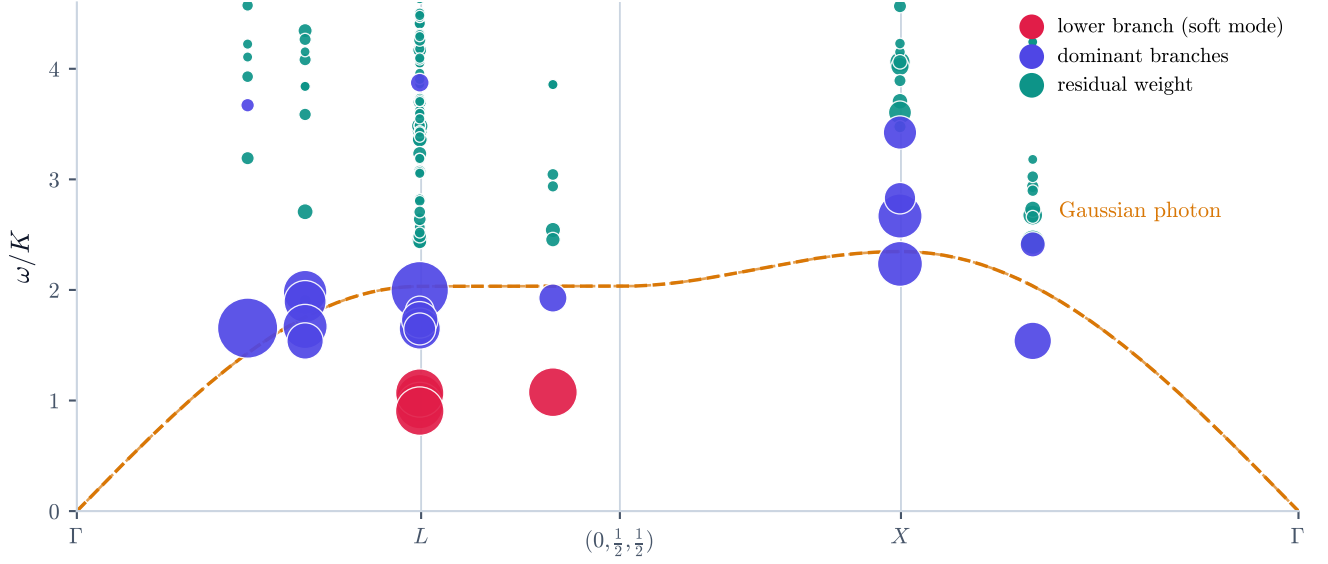


FIG. 1. **Order-one reconstruction at lattice momentum.** Operator-resolved S^z spectrum along $\Gamma \rightarrow L \rightarrow (0, \frac{1}{2}, \frac{1}{2}) \rightarrow X \rightarrow \Gamma$, combining all four diagonalized clusters; marker area is the fraction of the sublattice-summed weight carried by each level at its momentum. Red marks the lower branch where it reconstructs, indigo the dominant branches elsewhere, teal the residual weight. The dashed curve is the Gaussian photon with a single fitted energy scale. The long-wavelength response is photon-like; near L , and again at $(\frac{1}{6}, \frac{1}{2}, \frac{1}{2})$ on the 72-site cluster, the lower branch collapses to roughly half the Gaussian energy and absorbs half of the response.

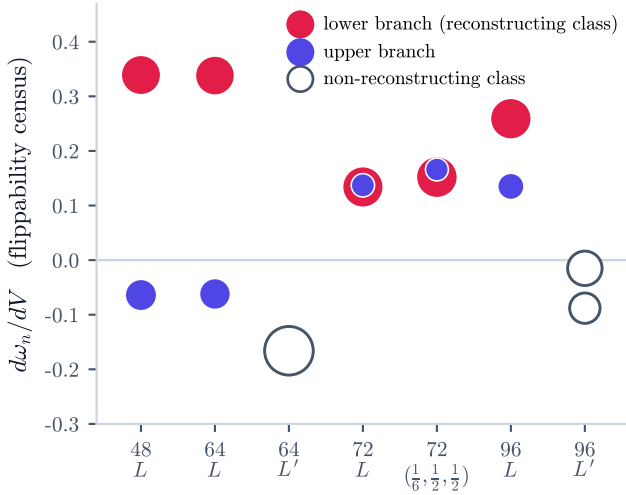


FIG. 2. **Microscopic fingerprint.** Flippability census [Eq. (2)] of the dominant levels. The robust statement is the within-cluster contrast: on the anisotropic 64- and 96-site clusters the lower branch of the reconstructing L class is positive while the dominant levels of the nonreconstructing class (open symbols) at the same $|\mathbf{q}|$ are negative. The upper branch's sign is not itself a diagnostic; it is negative at 48 and 64 and positive at 72 and 96.

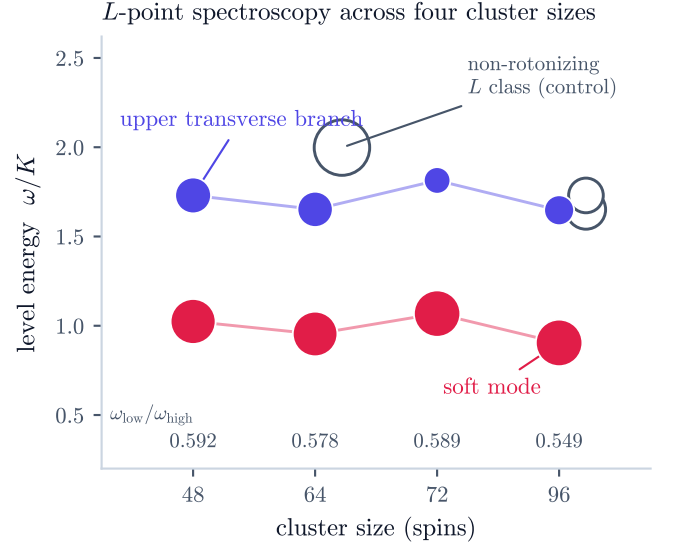


FIG. 3. **Finite-size stability and internal controls.** Filled symbols: lower and upper branches of the reconstructing L class. Open symbols: dominant levels of the inequivalent nonreconstructing class on the anisotropic 64- and 96-site clusters. Numbers give $\omega_{\text{low}}/\omega_{\text{high}}$. The lower branch retains approximately half of the sublattice-summed weight at every size.

commutator of $S^z(\mathbf{q})$ with H , is

$$f_1(\mathbf{q}) = \frac{K}{2N} \sum_p \langle W_p + W_p^\dagger \rangle \sum_\mu |g_p^\mu(\mathbf{q})|^2, \quad (3)$$

where $g_p^\mu(\mathbf{q})$ is the alternating lattice curl of the sublattice-resolved electric-field wave around plaquette p . Equation (3) has the form of a Maxwell magnetic energy: a purely geometric factor, the squared curl of the imposed wave, weighted by the measured ground-state ring coherence. It is the oscillator strength—the energy cost of creating the electric-field wave—and it contains no fitted parameter.

Their ratio f_1/S is the variational energy of the single-mode state $S^z(\mathbf{q})|0\rangle$, and equivalently the centroid of the spectrum that state creates. This is Feynman’s construction for the helium roton [22]: a collective excitation is cheap where the probe couples weakly and the static correlations are already strong. Two features make it useful here. It is an upper bound on the lowest excitation in the channel, so a small value certifies softness; and because both factors come from the ground state, it can be evaluated *in advance* and tested as a forward prediction. A low centroid does not by itself prove a sharp pole—that is what the diagonalizations supply—but it says where to look.

It succeeds three times. On 64 sites the two inequivalent L classes have $S = 0.369$ and 0.227 ; the lower-ratio class develops the $0.954K$ level with 50.0% of the weight, while the other stays dominated by a $1.998K$ level with 77%. On 96 sites, with $S = 0.355$ and 0.241 , the test repeats. On 72 sites the ratio at L and at $(\frac{1}{6}, \frac{1}{2}, \frac{1}{2})$ differ by 0.9%, and the two soft levels then appear within 0.7%, at $1.068K$ and $1.075K$ with 54% and 55% of their respective weights. The ratio does not predict absolute pole energies—the centroid also carries the high-energy weight—but it identifies which momenta reconstruct.

Both factors conspire at L . Geometrically, for $\mathbf{q} \parallel [111]$ one family of hexagons lies in the wavefronts of the electric-field oscillation and its alternating curl vanishes identically: the family-resolved geometric weights are $(96, 96, 96, 0)$ at L against $(96, 96, 96, 96)$ at X , reducing f_1 . Geometry alone is insufficient, since the same cancellation occurs at shorter wavelength along $[111]$ where nothing anomalous happens. The second ingredient is dynamical: at 48 sites $S(L) = 0.370$, whereas the equal-amplitude superposition over the same configurations gives 0.297 and classical loop Monte Carlo on a 2048-site ice manifold is flat among zone-boundary momenta. The enhancement is made by the interacting quantum amplitudes, not by the ice constraint—which is what distinguishes this from a Rokhsar–Kivelson resonon, whose softening follows directly from classical correlations.

Because f_1/S needs only the ground state, it can be evaluated far beyond exact diagonalization. The ring Hamiltonian is stoquastic in the ice basis, permitting sign-free Green’s-function Monte Carlo with forward

walking, benchmarked against the exact 48-site $S(L)$ to every digit. On complete allowed momentum grids of 864 and 2048 spins the ratio at L is $2.142(10)$ and $2.179(19)$, against $2.660(10)$ and $2.713(21)$ at X and 2.72 – 2.78 at W , K , and U : an L -centered minimum 20–25% deep that drifts by 1.7% between the two sizes (Fig. 4). The softness extends along a shallow $(t, \frac{1}{2}, \frac{1}{2})$ valley, which is why the 72-site cluster finds a second reconstructed momentum on that line.

Interpretation.—The phenomenology is that of a roton in Feynman’s sense: a weight-dominant collective excitation at finite momentum, located by the ratio of oscillator strength to static correlation, and lowered further by correlated relaxation. That the designation belongs to the lower branch is measured rather than assumed. The single-mode state built from $S^z(\mathbf{q}_L)|0\rangle$ overlaps the lower-branch multiplet with squared amplitude 0.88, 0.87, and 0.86 on the 48-, 64-, and 72-site clusters, coupling to the upper branch only through the orthogonal transverse channel: the Feynman construction produces this branch, not the other. Relaxing that state within the exact spectrum recovers 28–32% of its energy, a size-stable amount, through a strongly family-selective reorganization of ring coherence.

We state the limits precisely. The lattice modifies the geometry of the phenomenon: along the radial $[111]$ direction the centroid rises from Γ to L and folds back by periodicity, so there is no helium-like radial dip; the minimum lives in the zone-boundary manifold, where the competing short-wavelength excitations are. The finite-size branch descends into the zone boundary while the large-system centroid develops its minimum there. Sharpness is established up to 96 spins; beyond that the Monte Carlo constrains the centroid, not the pole, and whether the level broadens into a narrow continuum in the thermodynamic limit is open. Finally, the low mode is not a multiparticle state of the free theory: it is odd under spin reversal, excluding even photon numbers, and at $1.02K$ it lies a factor of five below the three-photon threshold of $5.55K$ computed on the same torus—an exclusion valid for weakly renormalized constituents, which we do not extend further.

Experimental implications.—The signature is a two-feature zone-boundary response where a renormalized photon would give one. Following the $[111]$ direction outward, a single dispersing mode at small $\mathbf{q}$ should separate into two, the splitting growing from unresolvable to nearly a factor of two by the zone boundary, with the *lower* feature carrying two to three times the weight of the upper. Two aspects of this are scale-free and therefore testable without knowing K : the ratio of the two energies, 0.55–0.59, and the ratio of their weights. A single photon branch with a renormalized velocity, however strongly renormalized, produces neither. Because the reconstruction tracks f_1/S rather than the label L , the same signature should appear along the $(t, \frac{1}{2}, \frac{1}{2})$ line,

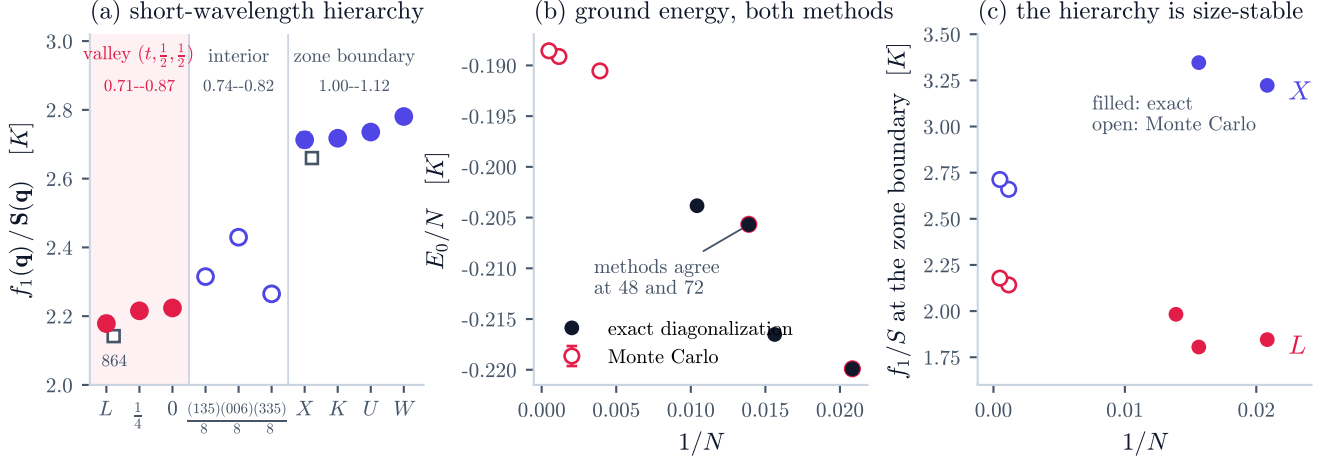


FIG. 4. **Large-system ground-state landscape.** (a) $f_1(\mathbf{q})/S(\mathbf{q})$ on the 2048-site supercell grouped by geometric role, with each group's $|\mathbf{q}|$ range indicated; interior stars lie below the boundary plateau only because the centroid is still rising with $|\mathbf{q}|$, and at matched $|\mathbf{q}| = 0.75$ the valley sits 9–10% below the interior star $(0, 0, \frac{3}{4})$. Open squares: 864-site values at shared momenta. (b) Ground-state energy from exact diagonalization and Monte Carlo, agreeing where both exist. (c) Size evolution at L and X across both methods.

as it does in our 72-site spectrum.

Two caveats bound this. A material-specific neutron cross section reweights the sublattice amplitudes coherently, so the theoretical 50–54% fraction should not be read as a universal neutron intensity; and longer ring exchanges, virtual spinons, and other microscopic terms can modify the zone-boundary spectrum of a real material. The robust prediction of the minimal model is the redistribution itself.

Discussion.—The emergent photon of quantum spin ice is nonperturbatively reconstructed at the lattice scale. Half of the longitudinal spectral weight moves into a mode near $1K$ whose scale-free ratio to its photon-like partner is stable across four clusters; the mode is microscopically distinguished by its flippability response, and its location follows a ground-state diagnostic that we verify predictively and carry to 2048 spins. What makes this possible is that the compact theory has no small parameter at short wavelength: the saturated electric field forbids increasing the amplitude of a field wave, so a finite-amplitude excitation must rearrange which bonds carry the two allowed values, and that rearrangement is cheapest where static correlations are strongest and the curl is weakest.

The broader implication is that the short-wavelength dynamics of a compact spin- $\frac{1}{2}$ gauge field are qualitatively richer than a weakly renormalized Maxwell photon. The obvious next questions are whether the reconstruction survives virtual spinons and longer ring exchanges in microscopic XXZ models, and whether the finite-size pole remains sharp in the thermodynamic limit—a question that requires dynamical rather than ground-state Monte Carlo.

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